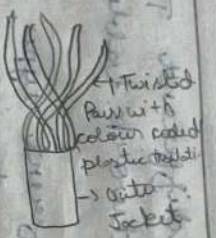
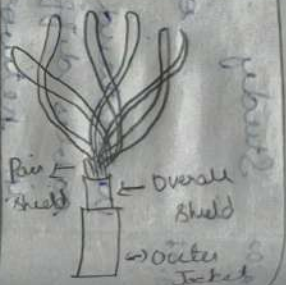
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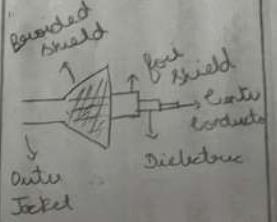
30.7.24

Study of different Types of Network cables

Understand different types of network cables different type of cables used in networking are:

1. Unshield Twisted Pair (UTP) cable
2. Shield Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre optic cable

Cable Type	Category	Maximum Data Transmission	Advantage / Disadvantage	Application / use	Image
UTP	Category 3	10 Mbps	Advantage: * Cheaper in cost * Easy to install as they have a smaller overall diameter Disadvantages * More prone to (EMI) Electromagnetic interference and noise	10 Base - T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5	upto 100 Mbps			
	Category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Advantage * Shielded * Faster than UTP * Less susceptible to noise and interference Disadvantages * Expensive * Greater installation effort	Gigabit Ethernet, 10 G Ethernet (55 m) Widely used in data centres Gigabit Ethernet, 10 G Ethernet (100 m)	
SSTP	Category 7	10 Gbps			

Coaxial cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantage : + High bandwidth. + Faster than UTP + Less susceptible to noise and interference * Versatile Disadvantages : + Limited distance + Cost * Size is bulky	Speed of signal is 500m television network high speed internet connections	
Fibre optic cable	Single mode Multi Mode	100 Gbps	Advantage : + High speed * High bandwith + High security Disadvantage : + Expensive + Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters	

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3.8.26

Result:

The types of network cables and making cross wired cable & straight through cable using clamping/crimping tool is studied